Galileo Optical Instrument Ltd. (B)[1]

After performing the preliminary analysis of Galileo's profit from the binocular order, Candace decided to obtain additional information for her model. She called the owner of the Taizhou supplier, Jia Yun, and recorded their conversation with the production manager's consent.

Reconstruction of Conversation with the Owner at the Taizhou Supplier

Candace: Thank you for your time. I hope you can answer a few questions regarding the disruption of your production plant. Please remember that what you tell me will not affect your contract with Galileo. We will greatly appreciate the information you provide as it will help us plan for the uncertainties ahead.

Owner: I will try my best to help. Please go ahead with your questions.

Candace: Thank you. In our previous conversation, you mentioned that it is most likely that some or all machines at the plant would be substantially damaged but fixable. Can you estimate the chance that this most likely scenario will happen?

Owner: Hmm, it is difficult to give an exact number. Based on my conversation with my engineers, I believe the chance of seeing this most likely scenario is slightly higher than half-half.

Candace: May I put 60% as the chance of this scenario, given your best guess?

Owner: Sure.

Candace: What about the other two scenarios, the optimistic scenario where the tools at the plant are still functional, and the pessimistic scenario where some or all machines are damaged beyond repair and need to be replaced? What chances do you put on those scenarios?

Owner: These numbers are even harder to estimate. I guess there is no reason for me to believe that the optimistic scenario should be either more or less likely than the pessimistic scenario. Other than that, I cannot tell you much.

---

[1] This case is developed by Yehua Wei and Peng Sun, Fuqua School of Business, Duke University in August 2020. The case is based on a consulting visit and various discussions with managers and the CEO of a manufacturing firm in China. To maintain the anonymity of the firm, all names used in the case are fictitious. In addition, the numbers presented the case are developed for classroom purposes and do not reflect actual business information of the firm.

Candace: Since we already agreed on the 60% estimate on the chance of the most likely scenario, does it sound reasonable to estimate 20% chance for the pessimistic scenario, and another 20% chance for the optimistic scenario?

Owner: You are really good with numbers! That sounds reasonable.

Candace: Thanks! My next question is about the projected production starting date in the *most likely scenario*. Previously, you told us that the production will start in five weeks…

Owner: As I said, that is the projected starting date. I don't know exactly how it is going to turn out.

Candace: Of course. Am I right to assume that in the *most likely scenario*, the median starting date is five weeks?

Owner: Yes, that sounds about right.

Candace: To better help us, can you estimate the earliest and latest plausible production starting dates in the *most likely scenario*? Rough estimate is fine, as I understand that there is some chance that the starting dates can be outside of your estimated range. I will use your numbers as the 10 and 90th percentiles, if you know what I mean.

Owner: Hmm, I would say the production will start no earlier than in three weeks, and at the latest in seven weeks.

Candace: That is very helpful. Can we do the same exercise for the *pessimistic and optimistic scenarios*? That is, I will need you to estimate the earliest and latest plausible production starting dates, as well as the median starting dates in those scenarios.

Owner: This is hard. Let me start with some very rough estimates first. For the *optimistic scenario*, I would guess the production starting date is four weeks, plus or minus one week; for the *pessimistic scenario*, my guess for the starting date is six weeks, plus or minus two weeks.

Candace: Thank you, I think I understand what you mean. Just to confirm, am I right to think that the median starting date is four weeks, with earlier and latest plausible dates being three and five weeks in the *optimistic scenario*?

Owner: Yes, and you can apply the same line of thought to the pessimistic scenario.

Candace: Great. I just have one more question. In all of these scenarios, when production resumes, would you achieve the original capacity of 1000 eyepieces per day?

Owner: I will have to ask my engineer for this one. I will let you know once I hear from him.

Candace: Thank you! Please ask him to provide the 10-50-90 range for the capacity, if he knows what that means. If not, may I arrange a call with him?

Owner: Sure, I will tell him.

Note: the numbers regarding each disruption scenario discussed between Jia Yun and Candace Shi are summarized as a table in Exhibit A.

Response from the engineer (forwarded by Jia Yun, the owner of the Taizhou supplier)
"In the case where tools at the plant are still functional [optimistic] or the case where some or all machines at the plant would be substantially damaged but fixable [most likely], we will be able to restore the plant to its original capacity when the production resumes. In the [pessimistic] case where some or all machines are damaged beyond repair and need to be replaced, we may not be able to restore the plant to its original capacity. My 10-50-90 estimation for the restored capacity would be 400, 500, and 800 eyepieces per day."

Salvage Value of the Eyepiece
Candace and Z.T. came up with two possible outcomes for the unused eyepieces. They estimated that there is a 20 percent chance Galileo will get future orders for similar binoculars requiring the same type of eyepieces. In that outcome, the unit salvage value of each eyepiece would be ¥5, the price they would have to pay if they buy from the Taizhou supplier. Otherwise, the unused eyepiece can be sold at a discount. Candace was willing to assume that the corresponding salvage value under that outcome is distributed uniformly with a range from ¥2 to ¥4.

Other Variables in the Model
After going over other key uncertainties in the model, Candace and Z.T. believed that it is fine to model them using their initial 10-50-90 estimations. Candace proposed to use piecewise-uniform distribution to model uncertainties whenever possible. Z.T. believes this is fine for all the uncertainties *except* RMB to U.S. dollar conversion rate, which should be modeled using the Normal distribution. Candace agreed.

Info on Taizhou supplier

			resume in (days)		
Scenarios	Scenario Label	Probability	Low	Base	High
Optimistic	1	20%	21	28	35
Most Likely	2	60%	21	35	49
Pessimistic	3	20%	28	42	56

Exhibit A: Days until Taizhou supplier resumes production.